

Deepak Mallampati

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Kent, OH - US-based, F-1 OPT (open to sponsorship)

PROFESSIONAL SUMMARY

Full-stack AI engineer who ships LLM-powered developer tools. Built and published a multi-provider browser extension (Manga Lens, Chrome Web Store) integrating four AI vision models, and React/TypeScript full-stack apps with structured prompt orchestration. Deep applied-LLM background - Retrieval-Augmented Generation (~35% retrieval-precision gain), agentic workflows (>30% hallucination reduction), embeddings and semantic retrieval - directly transferable to autocomplete and codebase retrieval. Master's in Computer Science (Kent State). Comfortable owning features end-to-end across TypeScript/React front ends, Python/FastAPI services, evaluation/benchmarks, and Docker deployment.

CORE COMPETENCIES

TypeScript • React • Node.js • LLM Application Development • RAG & Codebase Retrieval • Embeddings & Semantic Search • Agentic Workflows • Evaluation & Benchmarks • Prompt Engineering • Python / FastAPI • Docker

PROFESSIONAL EXPERIENCE

Progress Solutions Inc.

Jul 2025 - Present | USA

Jr. AI/ML Engineer Trainee - Healthcare Technology, AI Consulting

- Built **Retrieval-Augmented Generation** systems with recursive semantic chunking and transformer embeddings; improved contextual **retrieval precision ~35%** and reduced irrelevant retrieval **>30%** - the same retrieval/ranking discipline that drives codebase search and autocomplete quality.
- Developed **agentic LLM workflows** for multi-step queries with structured reasoning, tool discipline, and grounding rules; improved response stability **~25%**.
- Packaged ML/LLM inference as **FastAPI/Flask** RESTful services, containerized with **Docker**, with structured logging and load simulation; reduced post-deployment defects **~30%**.
- Shipped predictive ML pipelines (scikit-learn, XGBoost); improved recall **15-20%** on high-risk categories while holding precision **>90%**.
- Built evaluation audit trails and benchmark harnesses to measure retrieval and agent quality before release.

Suvidha Foundation

Jan 2022 - Mar 2022 | India

Machine Learning Engineer - Applied NLP, Video Analytics

- Built a transformer-based summarization system over 5,000+ sessions; hierarchical long-context summarization cut manual review time **60-70%**.
- Implemented **document Q&A with RAG** (semantic chunking + embedding retrieval); reduced hallucinated outputs **~30%**, raised answer accuracy **>85%**.
- Deployed the stack as a **Flask** API with a lightweight web interface for non-technical users.

Energy Solutions International (Emerson)

Jun 2022 - Apr 2023 | India

Database & DevOps Performance Engineer (Intern)

- Built **CI/CD pipelines (Jenkins)** for version-controlled SQL and stored-procedure deploys; reduced deployment errors **>30%**, improved release cycle time **~35-40%**.
- Tuned T-SQL stored procedures; cut query execution time **~20%** and retrieval latency **~25%**.

PROJECTS

Manga Lens - Chrome Extension for Real-Time AI Translation (shipped, Chrome Web Store)

Full-stack browser extension built independently - Manifest V3, content scripts, service workers, and **captureVisibleTab** for panel capture. Integrates **four AI vision providers** (Claude Sonnet, GPT-4o mini, GPT-4.1 Nano, Ollama/minicpm-v) behind a common interface, with per-provider payload handling, a seven-day translation cache, and per-domain selector configs for 29 sites.

TypeScript, Manifest V3, service workers, multi-LLM orchestration

Dream Decoder - Multimodal Creative Intelligence Platform

Full-stack **FastAPI + React/TypeScript/Vite/Tailwind** app coordinating multimodal APIs. Introduced intermediate **structured prompt transformation layers** - improved contextual alignment **~30%** and first-pass image success **~25-30%** over naïve direct-prompt orchestration.

React, TypeScript, Vite, FastAPI, structured prompt orchestration

Agentic Healthcare Claims & Fraud Risk Intelligence System

Multi-agent pipeline with **schema-validated JSON contracts between agents** to prevent cascading hallucinations; RAG-grounded validation against vector-indexed policy docs, ANN duplicate detection, and explainable, audit-ready reasoning traces.

Agents, schema contracts, RAG, vector search, evaluation

Geospatial Road Network Analysis & Community Detection

Parsed OpenStreetMap XML into a weighted **NetworkX** DiGraph; Dijkstra shortest-path routing, betweenness centrality on a 20k-node subgraph, Louvain community detection, and MST extraction. **~30%** runtime reduction via subgraph scoping - demonstrates algorithmic rigor on hard problems.

Python, NetworkX, graph algorithms

EDUCATION

Master's Degree, Computer Science - Kent State University, Ohio, USA

Aug 2023 - May 2025

SKILLS

Languages & Frameworks: TypeScript, React, Node.js, Python, FastAPI, Flask, SQL (PostgreSQL, T-SQL, SQLite), PHP

LLM & GenAI: LLM application development, RAG, agentic workflows, structured outputs, prompt engineering, evaluation pipelines, grounding, embeddings, vector/semantic search

ML Libraries: scikit-learn, XGBoost, PyTorch, Hugging Face Transformers, LangChain, LlamaIndex

Infra & DevOps: Docker, Jenkins, CI/CD, RESTful APIs, observability/logging, Grafana